

The Open University of Israel

Department of Mathematics and Computer Science

Guidelines for Writing Final Work and Thesis

**In the Framework of Master's Degree
Master of Science (M.Sc.) in Computer Science**

May 2021

Department of Mathematics and Computer Science

Computer Science Division

Head of Master's Program: Prof. Leonid Bernboim

Master's Departmental Committee: Dr. Ella Tzur - Committee Chair, Dr. Rivka Gadot -

Responsible for Final Works and Theses

Guidelines Coordination and Preparation: Dr. Maya Herman, Dr. Ella Tzur

Guidelines Update: Dr. Miriy Avigal, Dr. Rivka Gadot

Department of Mathematics and Computer Science Homepage:

<https://academic.openu.ac.il/cs/Pages/default.aspx>

Master's Degree in Computer Science Homepage:

<https://academic.openu.ac.il/CS/msc/Pages/default.aspx>

Contents

1	Part A: Introduction	3
1.1	Purpose of Final Work/Thesis	3
1.2	Stages of Preparing Final Work/Thesis	3
1.2.1	Selection of Topic and Supervisor	3
1.2.2	Proposal for Final Work/Thesis	3
1.2.3	Final Work/Thesis	4
1.3	Writing Proposal and/or Final Work/Thesis in English	4
2	Part B: Guidelines for Preparation, Writing, and Submission of Proposal for Final Work/Thesis	5
2.1	Purpose of the Proposal	5
2.2	Selection of Topic and Supervisor	5
2.3	Preparing the Proposal	5
2.3.1	Personal Details:	6
2.3.2	Project Details:	6
2.4	Ethics Committee	6
2.5	Proposal Submission	6
2.6	Proposal Review and Approval	7
3	Part C: Guidelines for Preparation, Writing, and Submission of Final Work/Thesis	8
3.1	Preparing Final Work/Thesis	8
3.2	Guidelines for Main Sections	9
3.3	Note:	10
3.4	Progress Report for Thesis and Final Work	10
3.5	Submission of Final Work/Thesis	10
3.6	Review of Final Work/Thesis	10
3.7	Presentation of Final Work/Thesis	11
3.8	Determination of Final Grade for Final Work/Thesis	11
3.9	Notes	11
4	Appendices	13
4.1	Appendix A: Rules for Recording Sources in Final Work/Thesis	13
4.2	Appendix B: Declaration Form on Independent Preparation of Final Work/Thesis	13
4.3	Appendix C: Editing Proposal and Final Work/Thesis	14

1 Part A: Introduction

1.1 Purpose of Final Work/Thesis

Students in the Master's program in Computer Science must submit a final work or thesis as part of their degree requirements.

The scope of a final work is equivalent to two seminar papers. The nature of the work is mostly theoretical. The purpose of the final work is to provide students with in-depth knowledge in a specific field and the ability to process current scientific material, the ability to write scientific work and present it.

The purpose of the thesis is to provide students with tools for innovative research work and to apply the knowledge they acquired during their studies. The thesis should reflect the students' research capabilities, their mastery of accepted scientific tools in the discipline, independent and original thinking, and ability to process scientific material.

Note: The primary responsibility for the work rests with you. In principle, we see the role of supervisors as advisors and guides, while students are responsible for all stages of the work: reading and studying, formulating the topic, taking the initiative to schedule meetings with the supervisor, writing and submitting the proposal and work, and the final presentation.

1.2 Stages of Preparing Final Work/Thesis

1.2.1 Selection of Topic and Supervisor

After completing 4 courses in the final work track or 3 courses in the thesis track, students must choose a supervisor and formulate a topic proposal for final work/thesis. This stage is important and requires thought and time. It is important to meet with several potential supervisors at this stage and carefully choose a topic that presents a challenge and interest. It is recommended to seek help from the person responsible for final works and theses in finding a supervisor and topic.

1.2.2 Proposal for Final Work/Thesis

- Reading appropriate background material and formulating the proposal
- Obtaining approval from the departmental ethics committee¹
- Approval of the proposal by the supervisor
- Submission of the proposal to the Master's departmental committee
- Review and approval of the proposal by the Master's committee

¹Only for works involving experiments with people.

These stages will be detailed in Part B.

Students must register for final work/thesis within two months of proposal approval and no later than six months from the completion of degree courses.

1.2.3 Final Work/Thesis

- Preparation of the work and approval by the supervisor
- Progress report
- Submission of the work for approval by the Master's departmental committee
- Review of the work by readers appointed by the Master's committee
- Presentation of the work before supervisors, readers, department faculty members, and students
- Determination of final grade

These stages will be detailed in Part C.

Completion of final work may take up to one year. Completion of a thesis may take up to two years. In certain cases, the Master's departmental committee may approve an extension of the submission deadline.

1.3 Writing Proposal and/or Final Work/Thesis in English

The proposal and/or final work/thesis should be in Hebrew. Students who wish to write them in English must submit a written request with justification to the Master's committee chair, along with the supervisor's recommendation, before writing the proposal. Only after receiving approval may students write the proposal and/or final work/thesis in English. A final work/thesis written in English must include a cover page, abstract, and table of contents in Hebrew.

2 Part B: Guidelines for Preparation, Writing, and Submission of Proposal for Final Work/Thesis

2.1 Purpose of the Proposal

The purpose of the proposal is to formulate in an organized manner a topic and direction for the work that meets academic standards. The proposal should be written in a way that demonstrates familiarity with the subject, knowledge of possible information sources, and the formulation of a worthy work plan that will lead to writing work at an appropriate level and within a reasonable time.

We recommend submitting the final work proposal before registering for the last two courses, as the supervisor may require studying certain courses in the field of the final work/thesis. Students who wish to submit a proposal for final work/thesis are recommended to take a seminar in the field of the final work/thesis before formulating the proposal. Only after successfully meeting the academic requirements of the seminar should they begin preparing the proposal for final work/thesis. After approval of the proposal for final work/thesis, work can begin.

2.2 Selection of Topic and Supervisor

The stage of selecting a supervisor and topic is one of the important stages in the process and it is recommended to invest time and thought in it. The person responsible for final works and theses can help you at this stage. The process of selecting a topic involves reading material related to the topic. It is recommended to contact potential supervisors and seek their help in formulating and defining the topic. A list of supervisors and possible topics that each of them offers can be found on the homepage of final works and theses for Master's degree in Computer Science at the link: https://academic.openu.ac.il/cs/msc/pages/thesis_final_work.aspx. You can also find examples of works done in the past on this site.

2.3 Preparing the Proposal

After selecting the topic and supervisor, a detailed proposal of objectives and work methods should be submitted, up to 5 pages in scope. At this stage, relevant material for the work should be collected and a general direction should be formulated. The preparation of the proposal is done under the guidance and approval of the supervisor.

It is important that the proposal be submitted in an appropriate manner both from the content aspect and from the formal aspect (see Appendix C). Care should also be taken for proper linguistic editing and style. Also, care should be taken for concise and accurate formulation and logical connection between paragraphs.

The proposal should include date, personal details and project details as detailed below:

2.3.1 Personal Details:

Student name; Student ID; Address; Phone; Email

2.3.2 Project Details:

- **Work name**
- **Supervisor name**
- **Work objective** (a few lines)
- **Background with reference to sources** (about one page)
- **Work description** (up to one page)
- **Work importance** (a few lines)
- **Methods for work execution** (what types of activities will be performed - reading, analysis, programming, etc. Tools should be detailed - software, etc. that will be used)
- **Work execution timeline** (detailed at monthly level)
- **Initial source list** (the proposal body should reference sources. Sources should be written according to accepted rules in professional literature, as detailed in Appendix A)

2.4 Ethics Committee

In every final work/thesis involving experiments with people, an approval from the departmental ethics committee must be attached to the proposal. If the work is done in collaboration with another institution, such as a university, school, or hospital, approval from the other institution must be added.

Details about the ethics committee, the referral process, and the referral form to the committee are available on the website: https://academic.openu.ac.il/cs/msc/pages/thesis_final_work.aspx

2.5 Proposal Submission

The proposal for final work/thesis should be submitted for supervisor approval, along with the accompanying page for final work/thesis proposal, which appears on the Master's website. The supervisor will forward the approved proposal along with the accompanying page to the Master's committee.

2.6 Proposal Review and Approval

The Master's committee will review the proposal and consult with additional readers for opinions as needed. The committee may decide in coordination with the supervisor one of these decisions:

- a. Approve the proposal
- b. Require the student to make corrections approved by the supervisor only
- c. Require the student to make corrections and then forward the proposal to additional readers for further reading
- d. Reject the proposal

The committee's decision will be communicated directly to the student by the Master's committee, with a copy to the supervisor.

3 Part C: Guidelines for Preparation, Writing, and Submission of Final Work/Thesis

3.1 Preparing Final Work/Thesis

Students will prepare the final work/thesis under the guidance of supervisors.

Students will ensure proper and appropriate linguistic editing of all parts of the final work/thesis, according to accepted rules of grammar, spelling, and citation in professional literature. Also, care should be taken for concise and accurate formulation and logical connection between paragraphs.

Final work/thesis should include these sections:

- Cover
- Title page
- Acknowledgments page (optional)
- Table of contents
- List of tables and figures
- Abstract
- Introduction
- Literature review
- Work objectives
- Work methods
- Presentation of results and their analysis
- Discussion and conclusions
- Summary and suggestions for further research directions
- List of sources
- Appendices (optional)
- Abstract (in English)*
- Table of contents (in English)*
- Acknowledgments page (in English - optional)*

- Back cover page (in English)*

Note: Requirements, changes, and additional details will be coordinated with the supervisor as needed. * If the work is submitted in English, these sections should be included in Hebrew.

3.2 Guidelines for Main Sections

Title page - The work name should capture the work topic and content. The title page should include the institution name, work name, student name, supervisor name, and date. See guidelines and example on the Master's homepage.

Table of contents - Care should be taken for division into chapters, sub-chapters, and sections related in logical development sequence. Care should be taken for identity between chapter and section names appearing in the table of contents and the chapter and section names found in the work body.

List of tables and figures - Includes a numbered list of table and figure names and corresponding page numbers. Also, references should be made in the work body to tables and/or figures.

Abstract - The abstract length will be about one page. The abstract must be comprehensive. Generally, the abstract should not include references to sources.

The abstract can include, for example: problem definition, summary of theoretical framework, main findings, scientific and applied conclusions.

Introduction - The introduction length will be one to two pages. The introduction will describe the work topic in a sharp and focused manner, and the work objective and its contribution in theoretical and/or practical aspects.

Literature review - The literature review should refer both to the broad topic of the work and to its focused topic. The review should be presented in a logical, critical, and comparative manner. The review can refer to early works in the field, but it must also refer to articles published in recent years in scientific journals.

Work objectives - Clear presentation of the problem investigated in the work. In presenting the problem, emphasis should be placed on its practical and/or theoretical value. There should be a connection between work objectives and the literature survey.

Research methods - Relates to tools and/or data collection process and/or validation and verification, analysis methods, etc.

Presentation of results and their analysis - Research results should be presented clearly and analyzed logically.

Discussion and conclusions - The discussion should include, among other things, interpretation of results, grounding of results in an appropriate theoretical framework, and an attempt to build a general theory. Also, care should be taken that work conclusions are based on the discussion.

Summary and suggestions for further research directions - The work summary will provide a comprehensive and complete picture of the entire work process. In the summary, the work importance should be explicitly stated, its contribution to deepening knowledge in the field and/or clarifying issues that were unclear before and/or how it suggests new research directions. Also, work limitations should be presented.

List of sources - Care should be taken for rules of source citation in the work body. One can choose one of the methods for recording and citing sources in one of the accepted styles and in coordination with the supervisor (see Appendix A). Every publication appearing in the source list should also appear in the work body and vice versa, every publication appearing in the work should also appear in the source list.

The source of every table and/or figure taken from an external source should be noted. The source in English should appear before the source in Hebrew.

Appendices - References should be made from the work body to appendices.

3.3 Note:

If a computer program was developed as part of the final work/thesis, it should be included in an appendix along with documentation, a sample run, and a CD of it.

3.4 Progress Report for Thesis and Final Work

Once every six months during the work, students must submit a periodic progress report to the supervisor. The supervisor will forward the report with comments and assessment of work progress to the Master's departmental committee. The report will be filed in the student's personal file and will be taken into account in the final assessment of the work.

3.5 Submission of Final Work/Thesis

The final work should be submitted for supervisor approval. After supervisors have approved the final work/thesis, students must submit it in one bound copy and one copy on a Word file for review to the Master's committee in Computer Science, along with the originality declaration form appearing in Appendix B.

3.6 Review of Final Work/Thesis

The Master's committee will appoint reader(s) who will read, comment on, and evaluate the final work/thesis. The readers may decide in coordination with the supervisor one of these decisions:

- a. Approve the work
- b. Require students to make corrections approved by the supervisor only

- c. Require students to make corrections and then return the work for further reading

The review process of the work will be confidential. Transfer of the additional reader's opinion in part or in full to the students' knowledge will be done anonymously or not anonymously according to the additional reader's request.

After review of the work, 3 updated copies (if changes were indeed made to the work) will be submitted to the Master's committee in Computer Science before the work presentation. Also, an abstract of about half a page will be submitted that will be sent with the lecture invitation. The invitation will be sent to all Computer Science faculty members and will also be published on the website.

3.7 Presentation of Final Work/Thesis

After approval of the work by the Master's committee, students must present their work in a lecture before department faculty members, supervisors, and examiners. The presentation duration is about one hour. It is advisable to base the presentation on a slideshow. It is important to consult with the supervisor regarding the number of slides in the presentation and regarding points worth emphasizing in the lecture.

3.8 Determination of Final Grade for Final Work/Thesis

The grade on the work will be calculated as follows:

- Weight of written work: 80%
- Weight of presentation: 20%

The Master's committee will forward the final grade of the work to supervisors and update the Academic Achievements Department accordingly. The Academic Achievements Department will send students a notice of the grade.

It is not possible to correct a grade after receiving the final grade.

To receive approval of eligibility for Master of Science (M.Sc.) degree in Computer Science, contact the Academic Achievements Department.

3.9 Notes

- Keep a copy of every document and form you submit
- Keep a copy of the work proposal and the work itself
- Keep the data and sources on which the work was based for a period of at least 3 years, for public review upon request

- Students who publish their work or part of it should note that the work was done at The Open University and is part of the requirements for receiving the Master of Science (M.Sc.) degree in Computer Science at The Open University

4 Appendices

4.1 Appendix A: Rules for Recording Sources in Final Work/Thesis

There are several styles of source recording and citation rules in writing academic work. Below is a detail of accepted styles and links to websites. It is recommended to consult with the supervisor for final work/thesis about the preferred style.

IEEE:

<http://www.ieee.org/documents/ieeecitationref.pdf>

ACM:

<http://www2.cs.ucy.ac.cy/~chryssis/specs/ACM-refguide.pdf>

APA:

<http://www.apastyle.org/elecref.html>

Chicago Style:

<http://www.bedfordstmartins.com/online/cite7.html>

4.2 Appendix B: Declaration Form on Independent Preparation of Final Work/Thesis

In final work/thesis, academic skills should be expressed, such as reading ability, understanding and analysis of scientific texts, integration of different types of knowledge, critical thinking, research planning and execution ability (in certain cases), and finally, fluent scientific writing.

The final work/thesis must be predominantly an independent creation of the students. To distinguish between their contribution and contributions of others, and to prevent harm to the "intellectual property" of writers on whom the work relies, writers must ensure citation of sources on which they relied.

In other words, the source of every statement or knowledge taken from others must be declared. This includes: direct quotes of statements or findings, ideas, opinions, and interpretations of other people. Deviation from citation and reference rules, not to mention taking without proper citation of parts of work or complete work of other writers, is interpreted severely as an attempt to present others' words as the writers' own words and is a serious violation of scientific ethics rules.

To prevent misunderstanding in this matter, we ask students to sign this declaration:

I _____ ID _____ hereby declare that the final work/thesis attached herewith is my own creation and was written according to accepted citation and reference rules in academia. Also, I declare that I am aware that submitting work in which many and/or significant and/or essential parts were copied from another work is a violation and that if it is discovered that I committed this violation, a complaint will be filed against me to the disciplinary committee of The Open University.

Signature: _____

Date: _____

4.3 Appendix C: Editing Proposal and Final Work/Thesis

The proposal/work should be printed, edited, and submitted according to academic writing rules and according to Open University guidelines as follows:

Font and Size:

- Font: David font, size 12
- Notes in smaller font

Line Spacing:

- In the final work/thesis body: One and a half lines
- In sources and footnotes: Single spacing

Margins:

- Right margins: 2-3 cm
- Page size: A4

Page Numbering: The first page of the abstract will be numbered 1. Pages should be numbered continuously until the end of the work, except for the part written in English. In this part, numbering should start from the first page of the abstract in English in Roman numerals.

Cover: The final work/thesis should be submitted bound. The covers (Hebrew and foreign) will also be printed on the back of the cover.